1. Motivation / Gap

Automatically discovered concepts in medical VLMs can be useful for interpretability, but their reliability is not guaranteed. Concepts may be ambiguous, too general, semantically inconsistent, or activated because of spurious correlations.

This gap is aligned with the project description, especially the lack of guarantees on semantic validity and stability of discovered concepts.

2. Main Research Question

Are automatically discovered concepts in medical images stable, meaningful, and reliable across similar samples?

Optional second question:

Can simple filtering or consistency checks help remove or flag unreliable concepts?

3. Proposed Setting

A simplified MedConcept-inspired setting adapted to chest X-ray images:

- use a pretrained vision or medical vision-language model, such as CLIP/BioMedCLIP, as the image and text encoder
- use a small feasible subset of MIMIC-CXR / MIMIC-CXR-JPG
- extract image embeddings from the chest X-ray images
- train a sparse autoencoder on these embeddings, following the SAE-based concept discovery idea from Discover-then-Name CBM [12]
- assign semantic names to the learned SAE neurons/features using cosine similarity with a small chest X-ray concept vocabulary
- analyze the reliability and consistency of the discovered concepts
- apply a simple filtering step to remove or flag unreliable concepts

We do **not** aim to reproduce the full MedConcept pipeline or train a large medical VLM. Instead, we focus on a smaller pipeline inspired by MedConcept, with our contribution centered on concept reliability evaluation and filtering.

4. Testing Ideas

Possible tests:

- **Consistency across similar images:** similar images or images with similar labels should activate similar concepts.

- **Over-activation test:** unreliable concepts may activate too often across many unrelated samples or labels.
- **Semantic alignment test:** assigned concept names should be semantically close to the learned SAE features and, when possible, coherent with image labels or reports.
- **Before/after filtering:** compare generated concept explanations before and after removing low-reliability concepts.

5. Possible Filtering Idea

Flag or remove concepts that:

- activate too broadly across unrelated samples
- are unstable across similar images
- have weak similarity with their assigned concept name
- are too generic to be clinically or semantically useful
- do not align well with available labels or report information, when available

6. Evaluation

Evaluate using:

- qualitative examples of generated concept explanations
- concept consistency score across similar samples
- semantic alignment score between SAE features and assigned concept names
- over-activation/generalness score
- number or proportion of filtered unreliable concepts
- comparison of explanations before vs after filtering

If feasible, we may also adapt MedConcept-style labels:

- **Aligned**
- **Unaligned**
- **Uncertain**

However, the main evaluation will focus on lightweight reliability and consistency metrics, so the project remains feasible.